

Customer Shopping Behavior Analysis

1. Project Summary

This project analyses consumer shopping behavior to assist management in identifying key sales drivers, customer demographics, and purchasing patterns across product categories and channels. Using Python for data preprocessing, SQL for targeted business problem-solving, and Power BI for interactive visualization, this report provides data-driven strategies to improve customer engagement and marketing effectiveness.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
 - Customer demographics (Age, Gender, Location, Subscription Status)
 - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
 - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Data Preparation & Exploratory Data Analysis (Python)

Data Overview & Structure

- Imported dataset (`customer_shopping_behavior.csv`) containing 3,900 rows and 18 attributes
- Primary data types identified: 1 float, 4 integers, and 13 categorical objects.

```
df.isnull().sum()
```

```
Customer ID      0
Age              0
Gender           0
Item Purchased   0
Category         0
Purchase Amount (USD)  0
Location         0
Size            0
Color           0
Season          0
Review Rating    37
Subscription Status  0
Shipping Type    0
Discount Applied  0
Promo Code Used  0
Previous Purchases  0
Payment Method   0
Frequency of Purchases  0
dtype: int64
```

```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))
```

Data Cleaning & Missing Value Handling

- **Missing Values:** Identified 37 missing values in the Review Rating column.
- **Imputation Strategy:** Imputed missing review ratings using median values grouped by product Category to preserve distribution integrity.

```
df.isnull().sum()
```

```
Customer ID      0
Age              0
Gender           0
Item Purchased   0
Category         0
Purchase Amount (USD)  0
Location         0
Size            0
Color           0
Season          0
Review Rating    37
Subscription Status  0
Shipping Type    0
Discount Applied  0
Promo Code Used  0
Previous Purchases  0
Payment Method   0
Frequency of Purchases  0
dtype: int64
```

```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))
```

- **Standardization:** Standardized column headers by converting text to lowercase, replacing spaces with underscores, and renaming purchase_amount_(usd) to purchase_amount.

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')
df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
```

Feature Engineering

- **Age Blanding (age_group):** Categorized customer ages into four equal bins using `pd.qcut`: Young Adult, Adult, Middle-aged, and Senior.

```
: labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
: df['age_group'] = pd.qcut(df['age'], q=4, labels=labels)

: df[['age', 'age_group']].head(10)
```

```
:   age  age_group
: ---  ---
0    55  Middle-aged
1    19  Young Adult
2    50  Middle-aged
3    21  Young Adult
4    45  Middle-aged
5    46  Middle-aged
6    63      Senior
7    27  Young Adult
8    26  Young Adult
9    57  Middle-aged
```

- **Frequency Standardizing (purchase_frequency_days):** Mapped string frequencies to numeric day values (e.g., 'Weekly' = 7, 'Fortnightly' = 14, 'Quarterly' = 90, 'Annually' = 365) to support quantitative analysis.

```

: # creating column purchase_frequency_days
frequency_mapping = {
    'Fortnightly':14,
    'Weekly':7,
    'Monthly':30,
    'Quarterly':90,
    'Bi-Weekly':14,
    'Annually':365,
    'Every 3 Months':90}
df['purchase_frequency_days']=df['frequency_of_purchases'].map(frequency_mapping)

```

4. Advanced Data Analysis (SQL SSMS)

The preprocessed dataset was imported into SQL Server Management Studio (SSMS) to answer ten core business queries:

Q1: Total revenue comparison between male and female customers.

Results			Messages	
	gender	revenue		
1	Male	157890		
2	Female	75191		

Q2: Analysis of customers who utilized discounts yet exceeded the average purchase amount.

Results			Messages	
	customer_id	purchase_amount		
1	2	64		
2	3	73		
3	4	90		
4	7	85		
5	9	97		
6	12	68		
7	13	72		
8	16	81		
9	20	90		
10	22	62		
11	24	88		
12	29	94		
13	32	79		
14	33	67		
15	35	91		
16	37	69		
17	40	60		
18	41	76		
19	43	100		
20	44	69		

Query executed successfully. FAIZ\SQLEXPRESS (16.0 RTM) FAIZ\faizk (74) customer_behaviour 00:00:00 863 rows

Q3: Top 5 products ranked by highest average review rating.

Results			Messages	
	item_purchased	avg_rew_date		
1	Gloves	3.86		
2	Sandals	3.84		
3	Boots	3.82		
4	Hat	3.8		
5	Handbag	3.78		

Q4: Average purchase amount comparison between Standard and Express shipping methods.

Results Messages		
	money_spend	shipping_type
1	58	Standard
2	60	Express

Q5: Revenue and average spend comparison between subscribers and non-subscribers.

118 %

No issues found

Results

Messages

	ttl_cust	avg_spend	total_revenue	subscription_status
1	1053	59	62645	Yes
2	2847	59	170436	No

Q6: Top 5 products with the highest percentage of discount usage.

Results Messages				
	item_purchased	no_of_item	discount_purchased	discount_perct
1	Hat	154	77	50
2	Coat	161	79	49
3	Sneakers	145	72	49
4	Sweater	164	79	48
5	Pants	171	81	47

Q7: Customer segmentation into New, Returning, and Loyal tiers based on previous purchase history.

Results Messages		
	customer_seg	total count
1	loyal	3116
2	returning	701
3	new	83

Q8: Top 3 most purchased products within each category.

Results Messages				
	category	item_purchased	cnt	rank
1	Accessories	Jewelry	171	1
2	Accessories	Belt	161	2
3	Accessories	Sunglasses	161	3
4	Clothing	Blouse	171	1
5	Clothing	Pants	171	2
6	Clothing	Shirt	169	3
7	Footwear	Sandals	160	1
8	Footwear	Shoes	150	2
9	Footwear	Sneakers	145	3
10	Outerwear	Jacket	163	1
11	Outerwear	Coat	161	2

Q9: Subscription likelihood analysis for repeat buyers (more than 5 previous purchases).

Results Messages		
	cnt	subscription_status
1	958	Yes
2	2518	No

Q10: What is the revenue contribution of each age group?

Results Messages		
	age_group	revenue
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Visual Insights & Dashboard (Power BI)



6. Business Recommendations

- **Targeted Marketing for Men & Young Adults:** Focus promotional campaigns toward male shoppers and young adults, as they drive the highest proportion of total revenue.
- **Subscription Conversion Push:** Offer incentive discounts to the 2.85K non-subscribers to improve long-term customer retention and recurring purchases.
- **Category Optimization:** Expand inventory and marketing efforts around high-performing categories like Clothing and Accessories, while bundling low-performing Outerwear to boost average order value.